

EQUIPMENT SPECIFICATION SHEET

Model: ESC-9500 Ultra-High Uniformity Electrostatic Chuck

System Class: 300mm Advanced Plasma Etch & Deposition Systems

Document ID: SPEC-ESC-2026-REV4

1. System Overview & Core Capabilities

The ESC-9500 is a high-performance, multi-zone electrostatic chuck designed for sub-2nm semiconductor manufacturing equipment. It provides active thermal control, high-clamping force stability, and chemical resistance during critical plasma etch and atomic layer deposition (ALD) processes.

- **Wafer Compatibility:** 300 mm (12-inch) standard silicon wafers
- **Clamping Technology:** Bipolar Coulombic / Johnsen-Rahbek hybrid design
- **Base Substrate Material:** High-purity Aluminum Nitride (AlN) ceramic
- **Top Surface Coating:** Yttria (Y₂O₃) plasma-resistant dielectric layer

2. Thermal Performance & Multi-Zone Control

- **Operating Temperature Range:** -20°C to +200°C
- **Thermal Heating Zones:** 32 independently controlled micro-zones (embedded resistive heaters)
- **Temperature Uniformity:** ±0.15°C across 300mm active wafer surface
- **Backside Helium Cooling:**
 - Dual-zone gas delivery (Inner/Outer)
 - Flow Rate Range: 0 to 50 sccm
 - Pressure Control Range: 2 to 30 Torr

3. Electrical & RF Specifications

- **DC Clamping Voltage:** ±500 V to ±2500 V (programmable bipolar)
- **De-clamping Time:** < 500 ms with integrated residual charge dissipation
- **RF Power Capability:** Up to 8 kW continuous power at 13.56 MHz / 60 MHz dual-frequency bias
- **Leakage Current:** < 10 µA at maximum operational clamping voltage

4. Mechanical & Physical Interface

- **Overall Diameter:** 318 mm
- **Total Assembly Height:** 145 mm (including lower cooling plate assembly)
- **Mounting Interface:** Standard 8-bolt vacuum flange mount with O-ring seal groove
- **Lift Pin Integration:** 3-pin pneumatic lift actuator pass-through ports
- **Coolant Fluid Channels:** Integrated counter-flow loop for Galden / Fluorinert heat transfer fluid

5. Operating Environment & Reliability

- **Base Pressure Limit:** Ultra-High Vacuum (UHV) compliant down to 1×10^{-9} Torr
- **Mean Time Between Failures (MTBF):** > 50,000 RF plasma process hours
- **Cleanroom Rating:** ISO Class 1 compatible